



Accelerate Agentic AI Development with NVIDIA

Saurav Agarwal



Agenda

- Introduction to Agentic AI

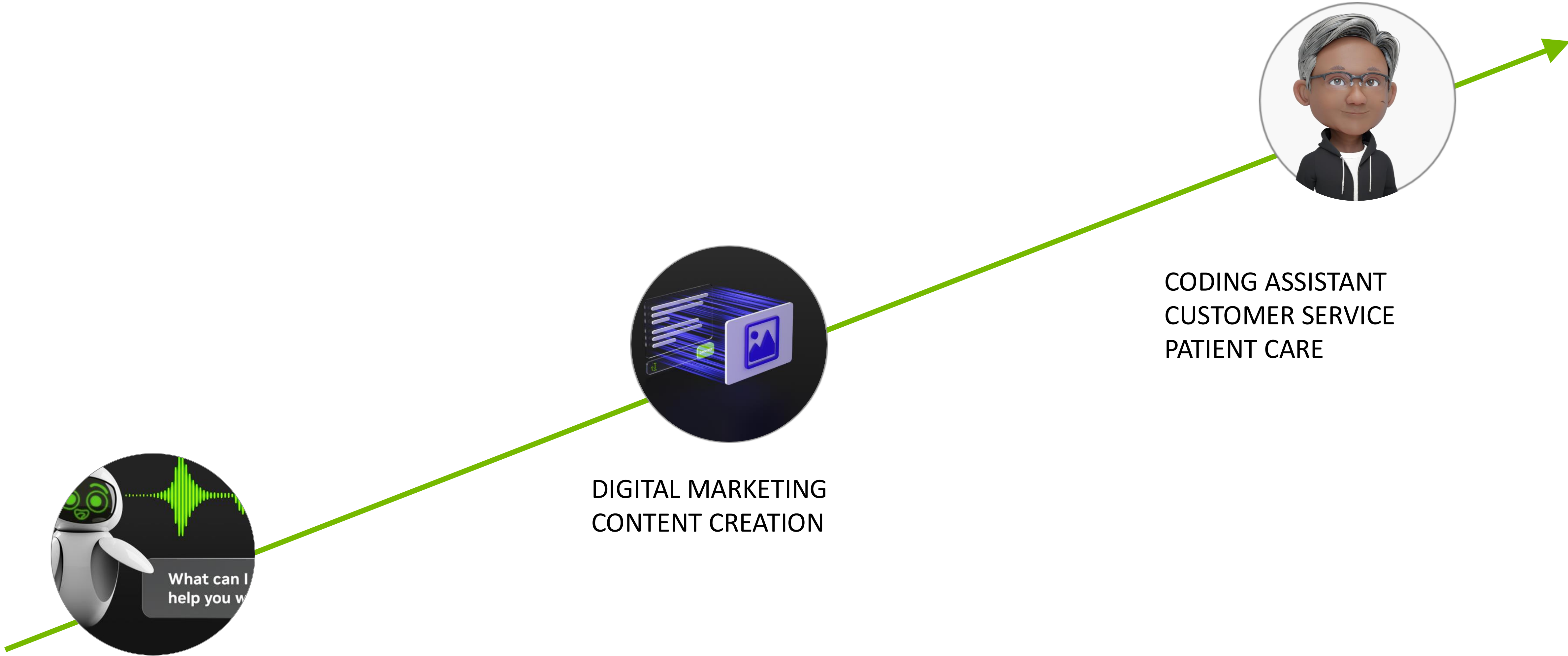
- NVIDIA Agent Intelligence Toolkit

- NVIDIA AI Blueprints

- Hands-on Demo

Evolution of AI

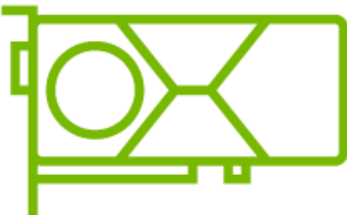
Agentic AI Enables More Powerful AI Applications



SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

DIGITAL MARKETING
CONTENT CREATION

CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE



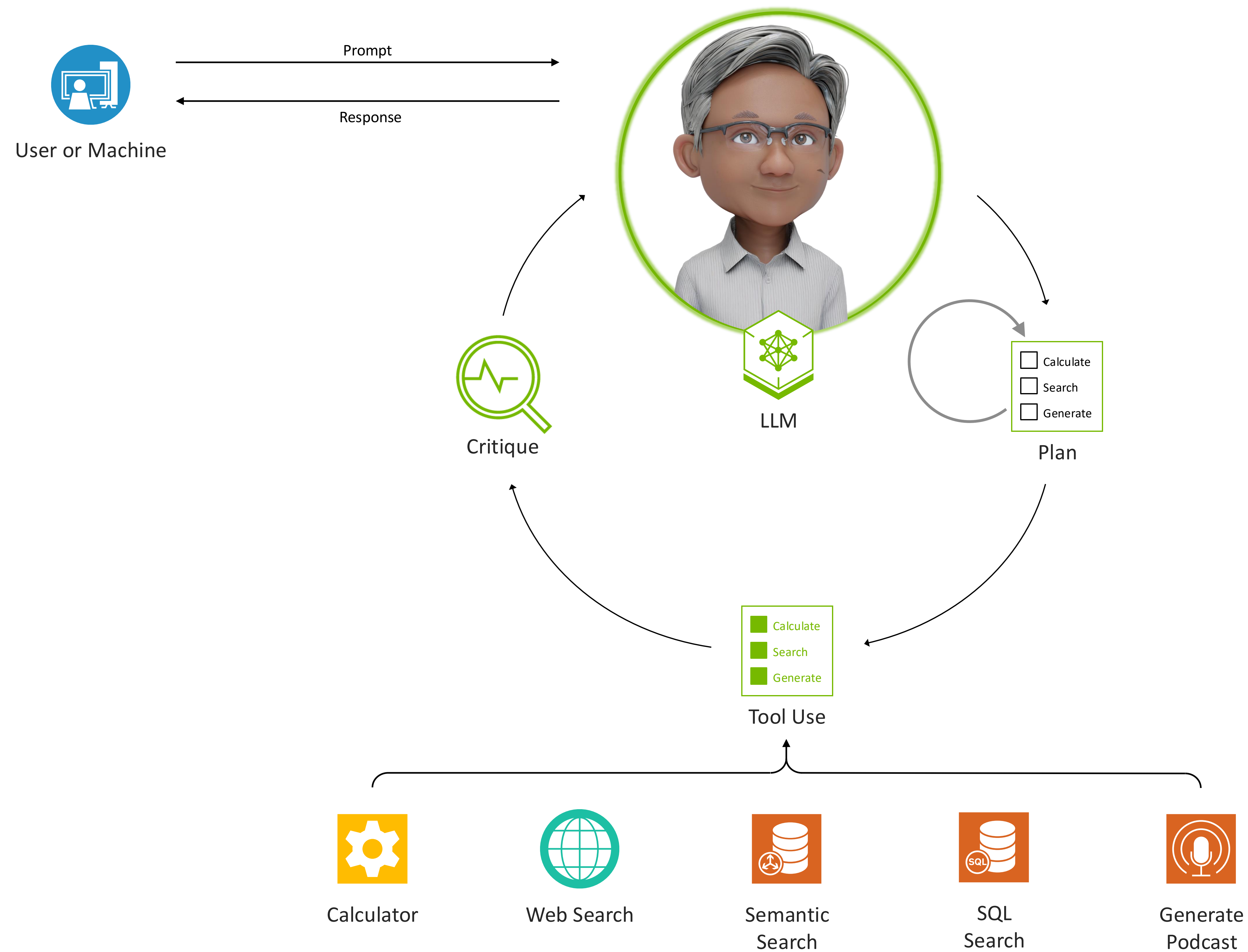
PERCEPTION AI

GENERATIVE AI

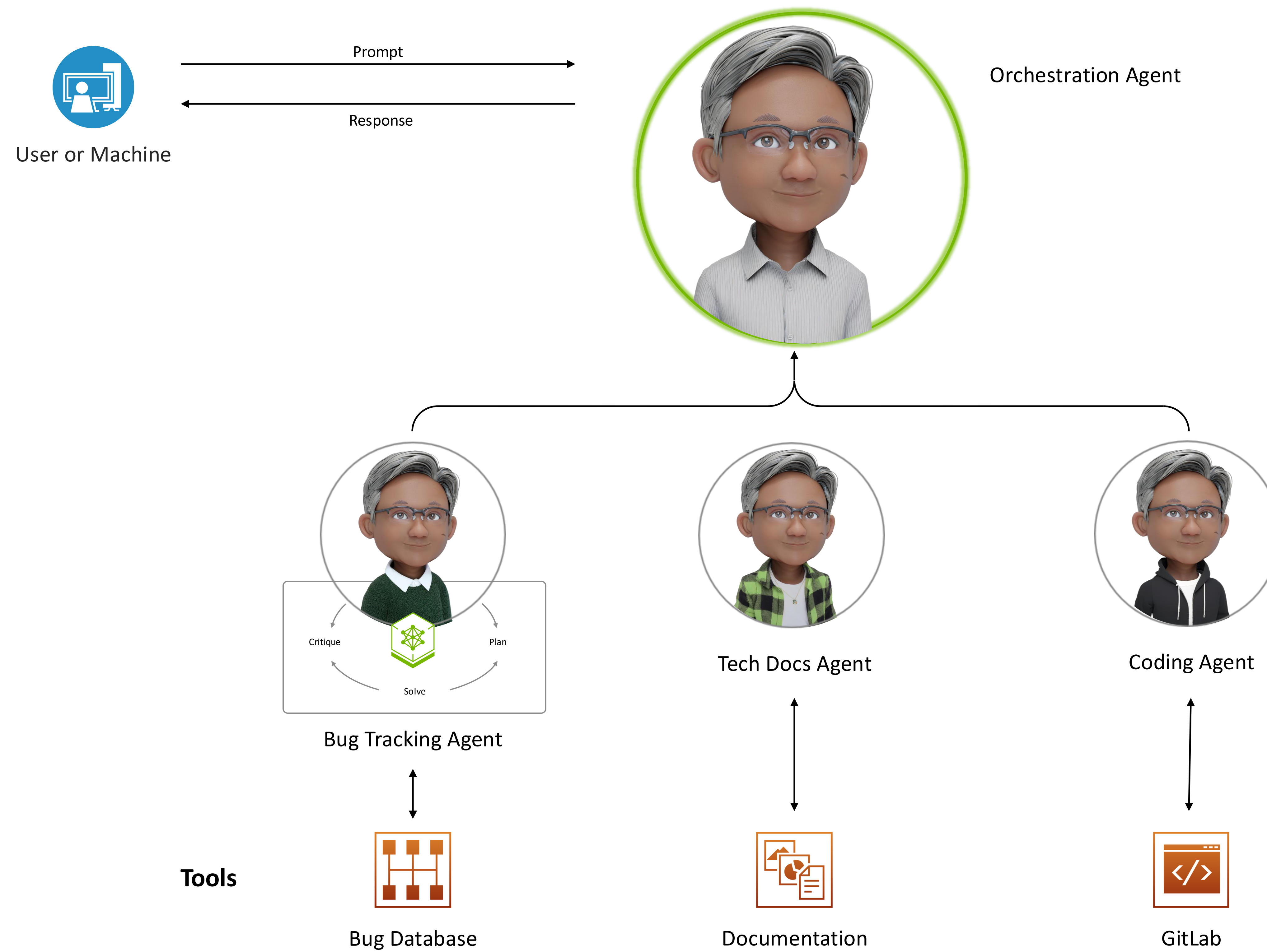
AGENTIC AI

How AI Agents Work

AI Agents solve problems and use tools to complete work



Specialized Agents Form Teams to Automate Work



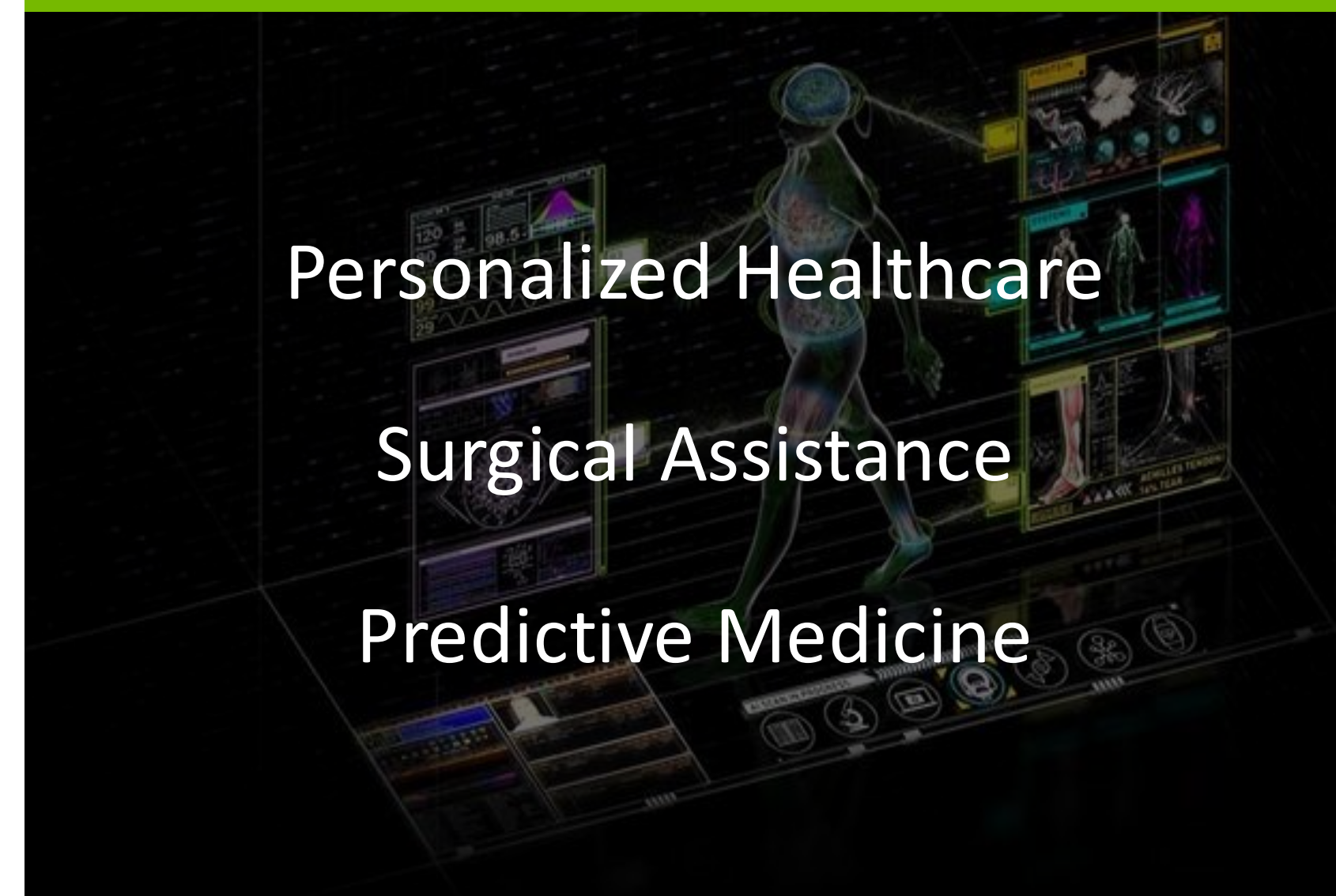
Key Agentic AI Use Cases by Industry

Every vertical has opportunities

Finance



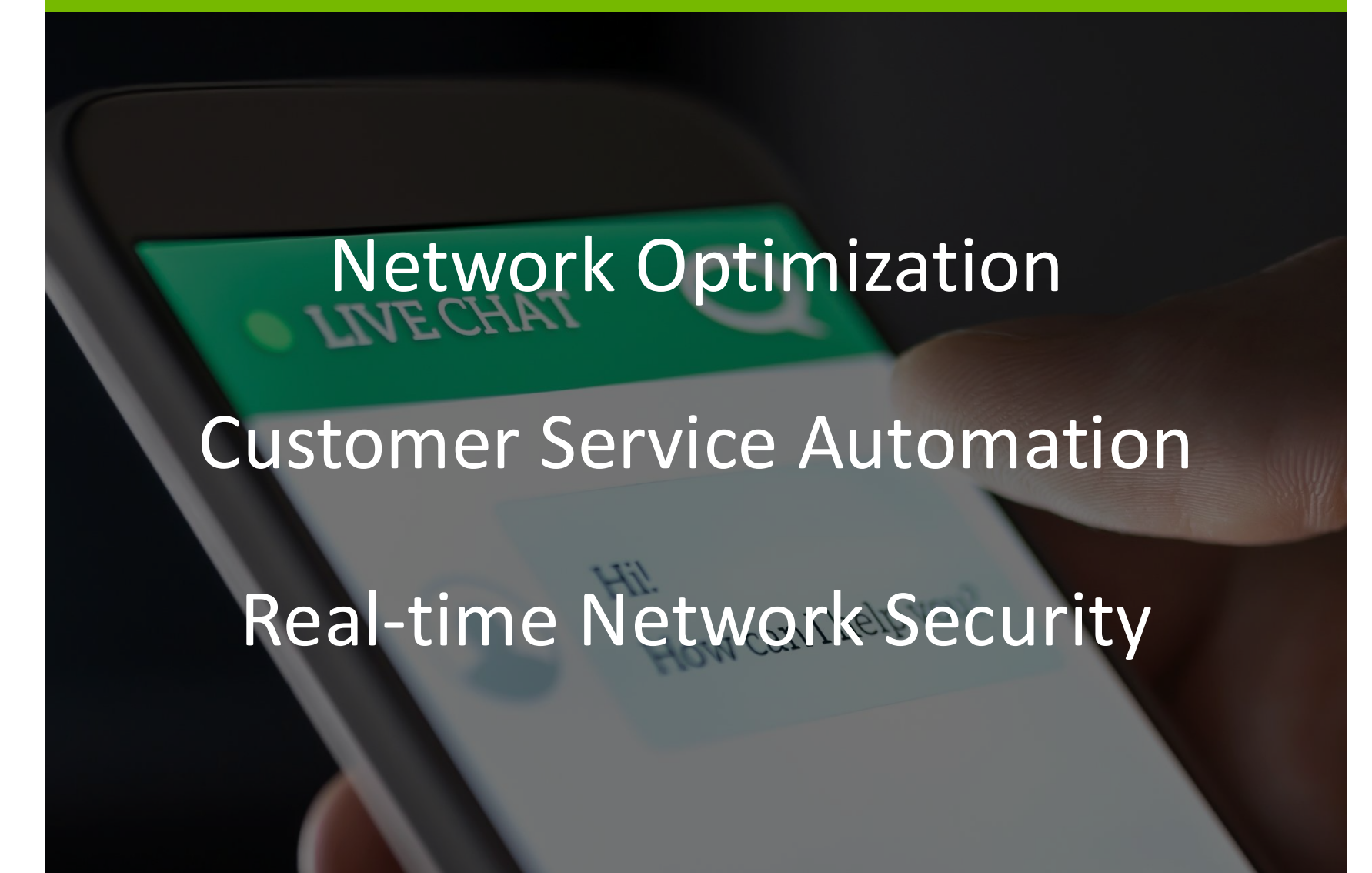
Healthcare



Retail



Telecommunications



Media & Entertainment



Manufacturing



Federal



Energy

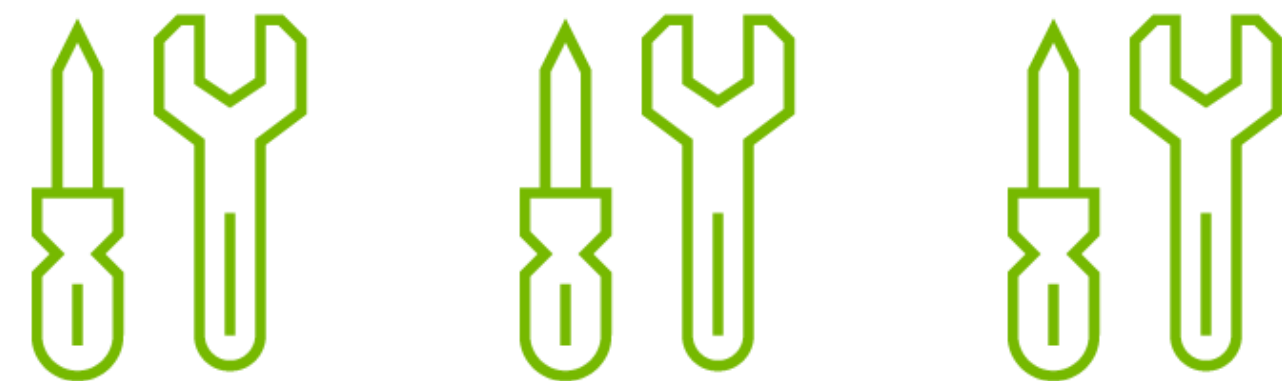


Building Agentic Systems is Complex

Agentic applications should integrate with existing tools, heterogeneous data, and perform reliably

Architectural Complexity

Many tools exist—but don't work together



Repeatability

Challenging to guarantee consistent results



Code Reuse

Fragmented solutions result in duplicate work



Performance

System-level accelerations require knowledge of the entire system



Learning & Adaptation

Continuous feedback and improvements



Safety, Security & Privacy

Preventing unwanted behaviors and protecting proprietary data



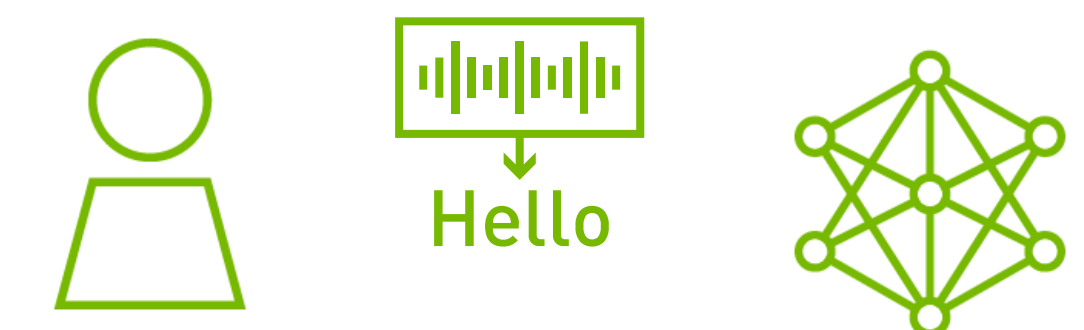
Enterprise Readiness

Evaluation system and telemetry



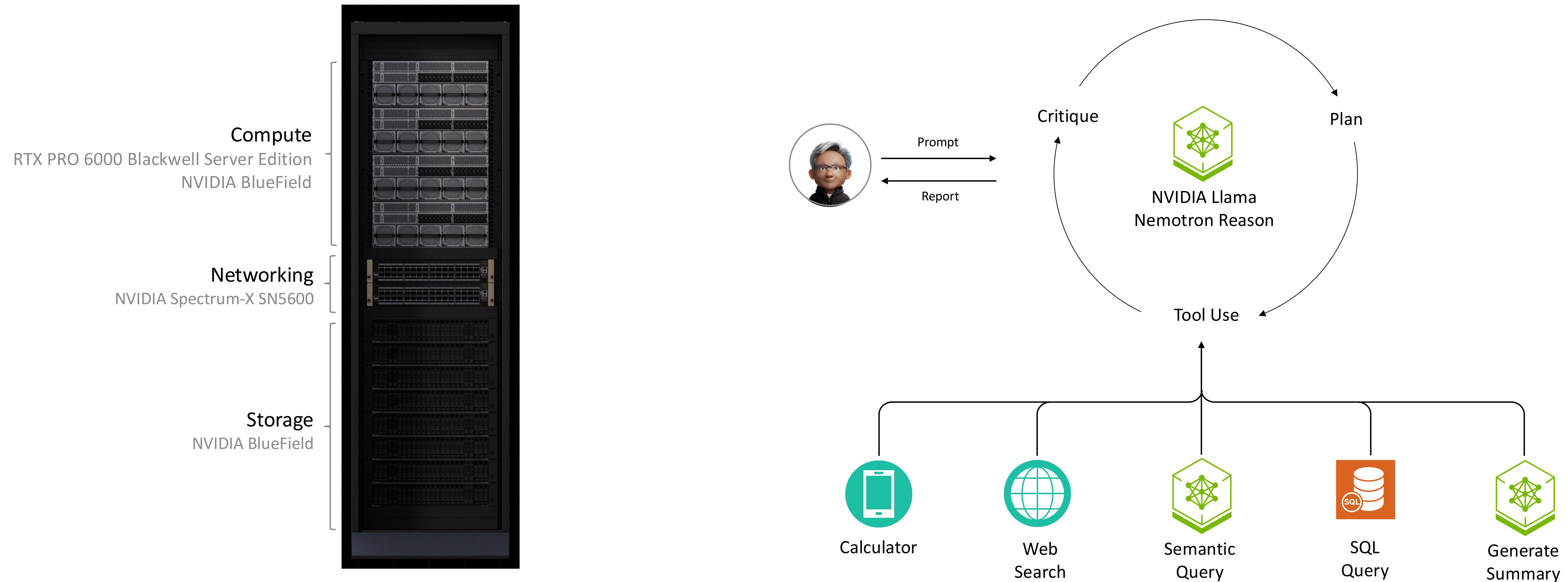
User-AI Interaction

Natural, effective communications and understanding



AIQ: AI Agent Interface to Enterprise Data Stores

AI Data Platform Reference Design



NVIDIA Agent Intelligence (AIQ) Toolkit

An open-source library for building enterprise-ready agentic systems

Profiling & Optimizations

- Fine-grained AI workflow telemetry collected can be used to implement agentic system accelerations.

Evaluation & Observability

- Evaluate system level accuracy
- Understand and debug inputs and outputs for each component in the AI workflow

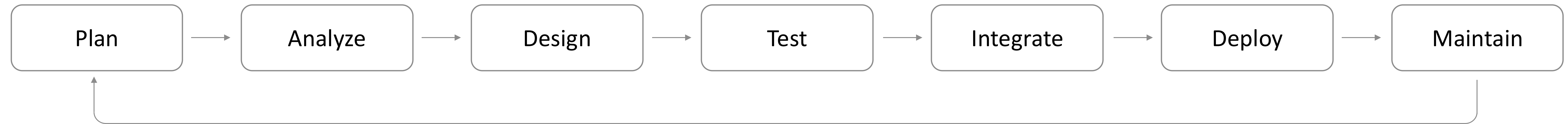
Agent Interoperability

- Universal descriptors for agents, tools, and workflows across frameworks
- Reusable Agent/Tool registry
- Workflow Configuration/Builder

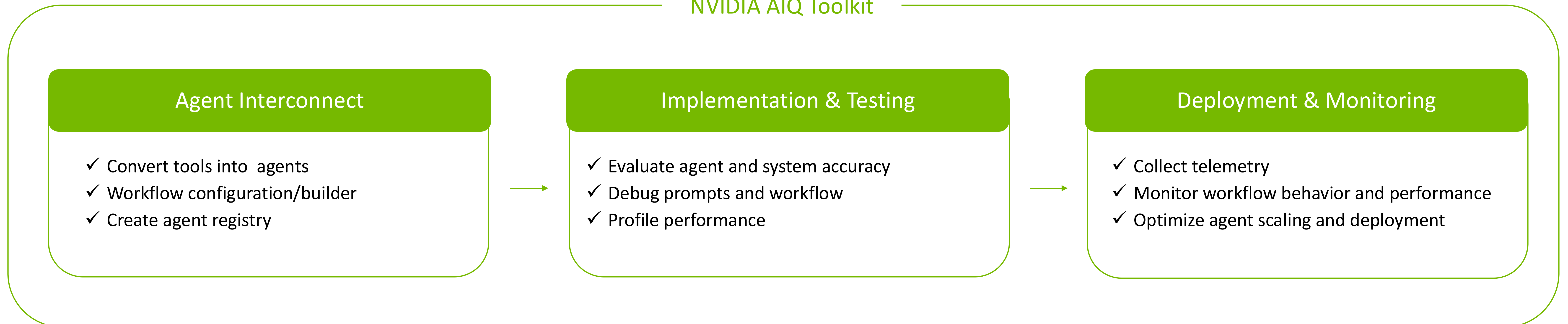


Enabling Software Development Lifecycle for AI Agents

NVIDIA AIQ Toolkit



NVIDIA AIQ Toolkit



NVIDIA AIQ Toolkit

Accelerate AI Agents and Streamline Agentic Workflow Optimization

SAVE TIME



Simplify the development of agentic systems

- Flexibly choose, and connect, agent frameworks best suited for each task
- Easily reuse existing and new RAG pipelines, different agentic workflows, and tools across your Enterprise
- Quickly elevate existing Gen AI workflows to agentic AI workflows

REDUCE COSTS



Accelerate agent responses—do more with what you have

- System level optimizations provide accelerated Agentic AI performance
- NVIDIA Agent Intelligence Toolkit collects telemetry that provides opportunities for optimization, driving efficiency for an agentic workflow.

IMPROVE BUSINESS OUTCOMES



Increase agentic system accuracy

- Evaluate agentic system response accuracy
- Understand and debug inputs and outputs for each component in the system
- Traceability and auditing of agent communications

Configuring a Heterogeneous AI Query Engine

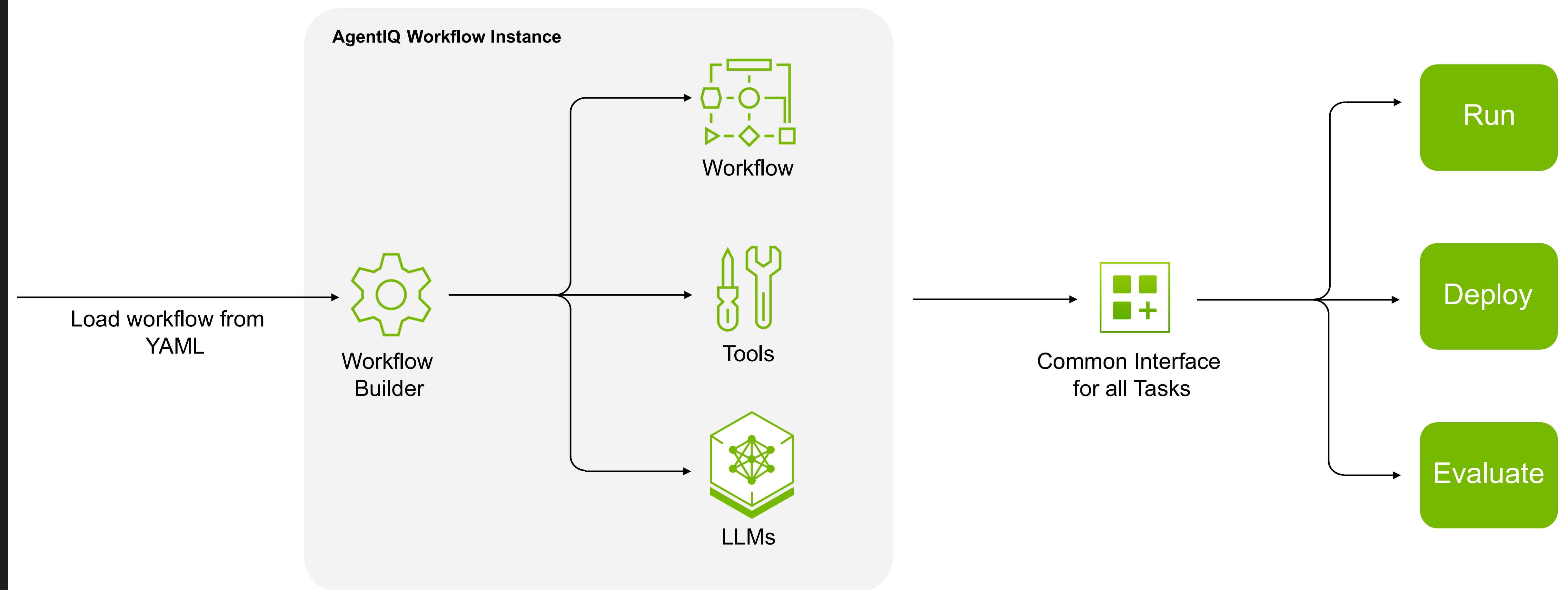
```
general:
  use_uvloop: true

functions:
  llama_index_rag_tool:
    _type: llama_index_rag
    llm_name: nim_llm
    embedding_name: nim_embedder
  langchain_rag_tool:
    _type: langchain_rag
    llm_name: nim_llm
    embedding_name: nim_embedder
  semantic_kernel_rag_tool:
    _type: semantic_kernel_rag
    llm_name: nim_llm
    embedding_name: nim_embedder
  react_agent:
    _type: tool_calling_agent
    tool_names:
      - llama_index_rag_tool
      - langchain_rag_tool
      - semantic_kernel_rag_tool
    llm_name: nim_llm

llms:
  nim_llm:
    _type: nim
    model_name: meta/llama-3.3-70b-instruct
    temperature: 0.0
    max_tokens: 1024
  nim_reasoning:
    _type: nim
    model_name: nvidia/llama-3.3-nemotron-49b-instructllama_nemotron
    temperature: 0.5
    max_tokens: 1024
    top_p: 1
    max_tokens: 1024

workflow:
  _type: reasoning_agent
  llm_name: r1_model
  augmented_fn: react_agent
  verbose: true
```

- **Reasoning Agent**
 - Plans the execution strategy using a Nemotron reasoning NIM
- **Tool Calling Agent**
 - Executes instructions generated by the Reasoning Agent using a Llama instruct model
- **Heterogenous RAG**
 - Each RAG engine exposed as a tool providing a heterogeneous AI query Engine.



Observability and Ease of Integration

AI-Q allows easy observability across popular agentic frameworks and observability tools

This notebook orchestrates a symphony of AI frameworks

- OpenAI for equation generation,
- LangChain for structured data processing
- LangGraph for workflow management,
- CrewAI for multi-agent collaboration

All unified through AgentIQ with Phoenix observability that visualizes the entire cognitive process in real-time.

The screenshot displays the Phoenix observability interface, showing a detailed trace for a 'react_agent' with a total latency of 26.07s. The trace is visualized as a tree structure, detailing the execution of various components and their associated latencies.

Trace Details:

- Trace Status: UNSET
- Latency: 26.07s

Trace Components (Latency):

- react_agent (26.07s)
 - meta/llama-3.1-70b-instruct (24403, 1.11s)
 - equation_generator (0, 0.54s)
 - equation_generator (0.54s)
 - meta/llama-3.1-70b-instruct (34944, 1.23s)
 - word_problem_generator (0, 1.57s)
 - word_problem_generator (1.56s)
 - meta/llama-3.1-70b-instruct (318, 1.55s)
 - meta/llama-3.1-70b-instruct (102941, 2.54s)
 - equation_solver (0, 2.87s)
 - equation_solver (2.86s)
 - meta/llama-3.1-70b-instruct (227, 1.53s)
 - meta/llama-3.1-70b-instruct (222, 1.23s)
 - meta/llama-3.1-70b-instruct (294489, 1.73s)

Input:

```
1 "Create a math problem about for pre-algebra students"
```

Output:

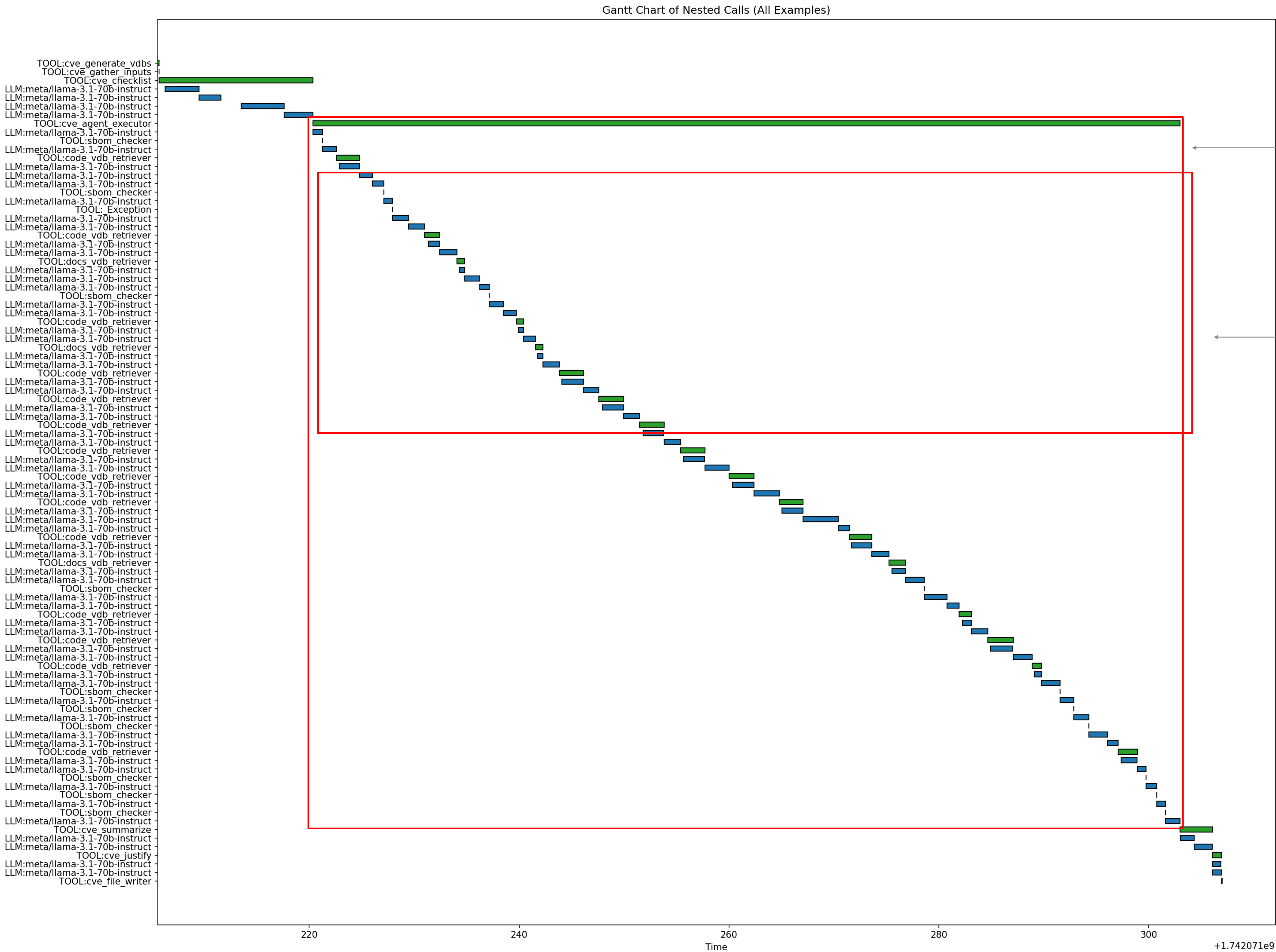
```
1 "The equation is 2x + 7 = 19. The word problem is: Tom has been saving money"
```

Metadata:

- Status: UNSET
- Start Time: 3/17/2025 00:29:22 AM
- End Time: 3/17/2025 00:29:48 AM
- Latency: 26.07s

Identifying Workflow Bottlenecks with the Profiler

Pre-optimization: Understand where agents spend most of their time



It appears that the Agentic Triage step is taking the longest time, at **82.62 seconds***

Most of the latency appears to be coming from the sequential execution of tools and agents* for each checklist item. **Could we do better?**

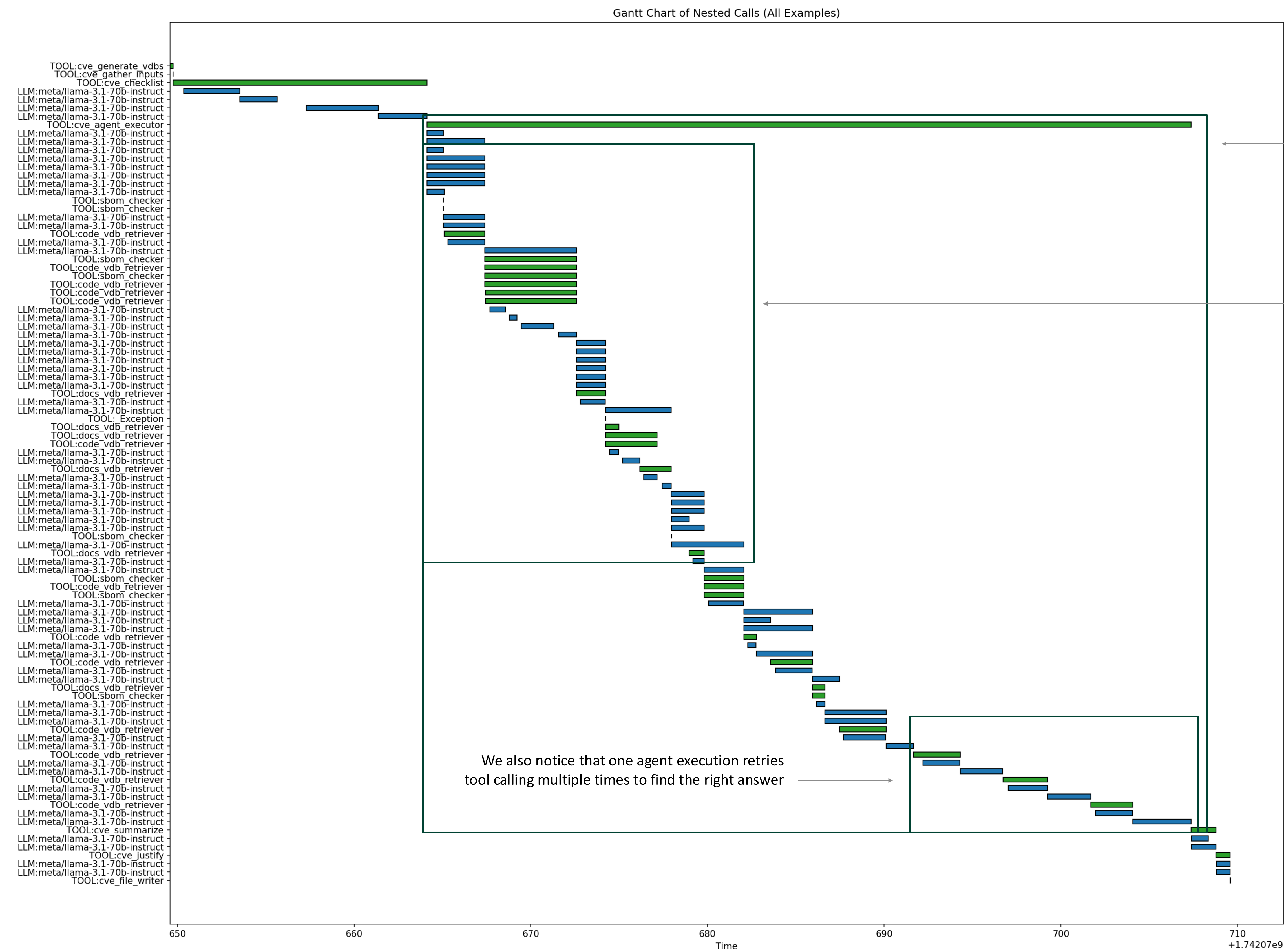
Key Ideas for Optimization:

- CVE's can be processed independently and instead of having to wait for one CVE to process before another does!
- Checklist items for each CVE can also be processed in parallel to reduce macro-latency.

* All metrics are extracted from the AgentIQ profiling report, not pictured here for brevity.

Identifying Workflow Bottlenecks with the Profiler

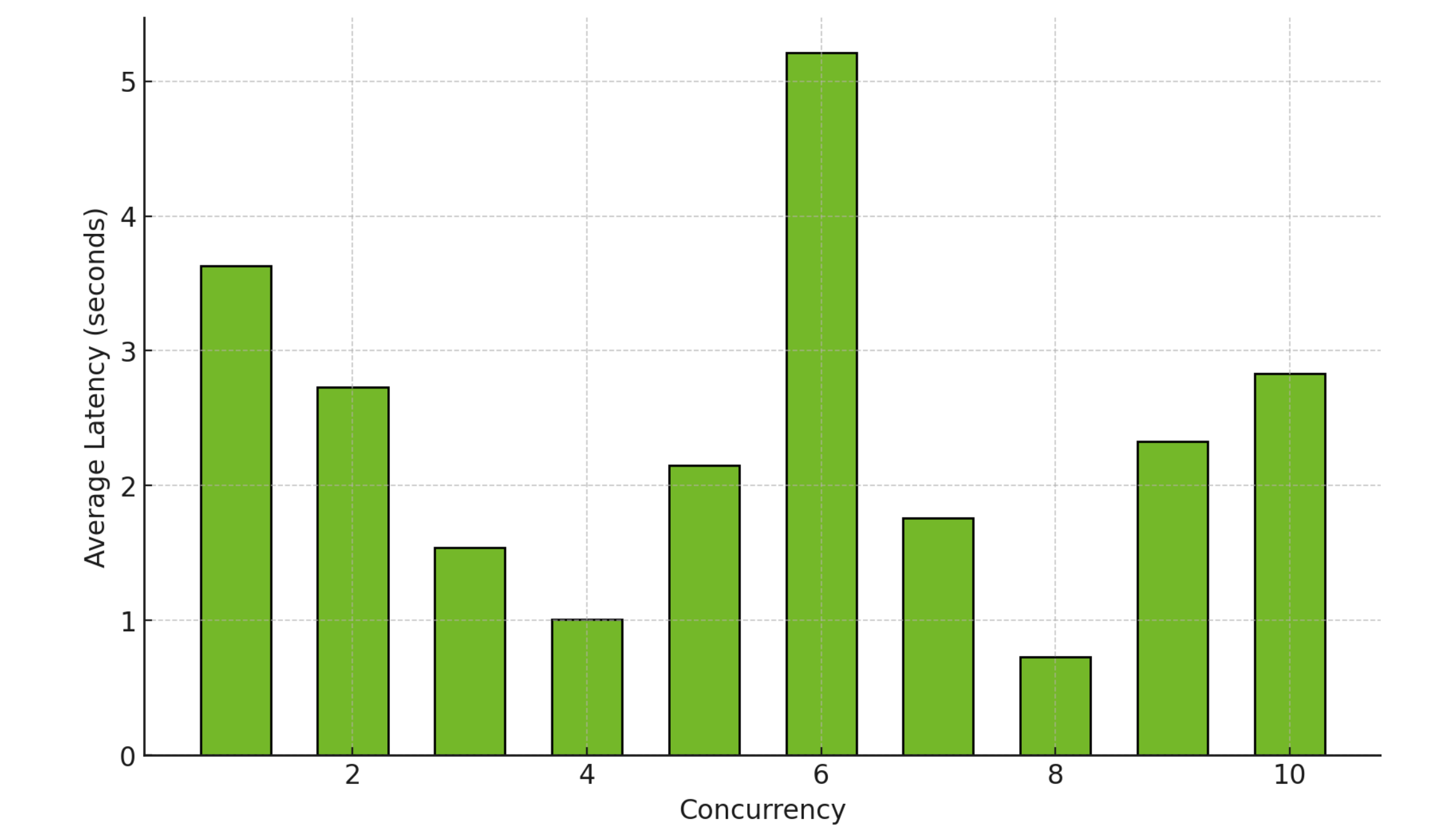
Targeted Optimization of Identified Bottlenecks



We notice a **2x reduction in latency by remediating identified bottlenecks** using optimization strategies. The total runtime is now 42.26 seconds.

Agents are executed in parallel for each CVE and checklist item. **Reducing end-to-end latency while saturating the GPU.** This will ultimately reduce TCO.

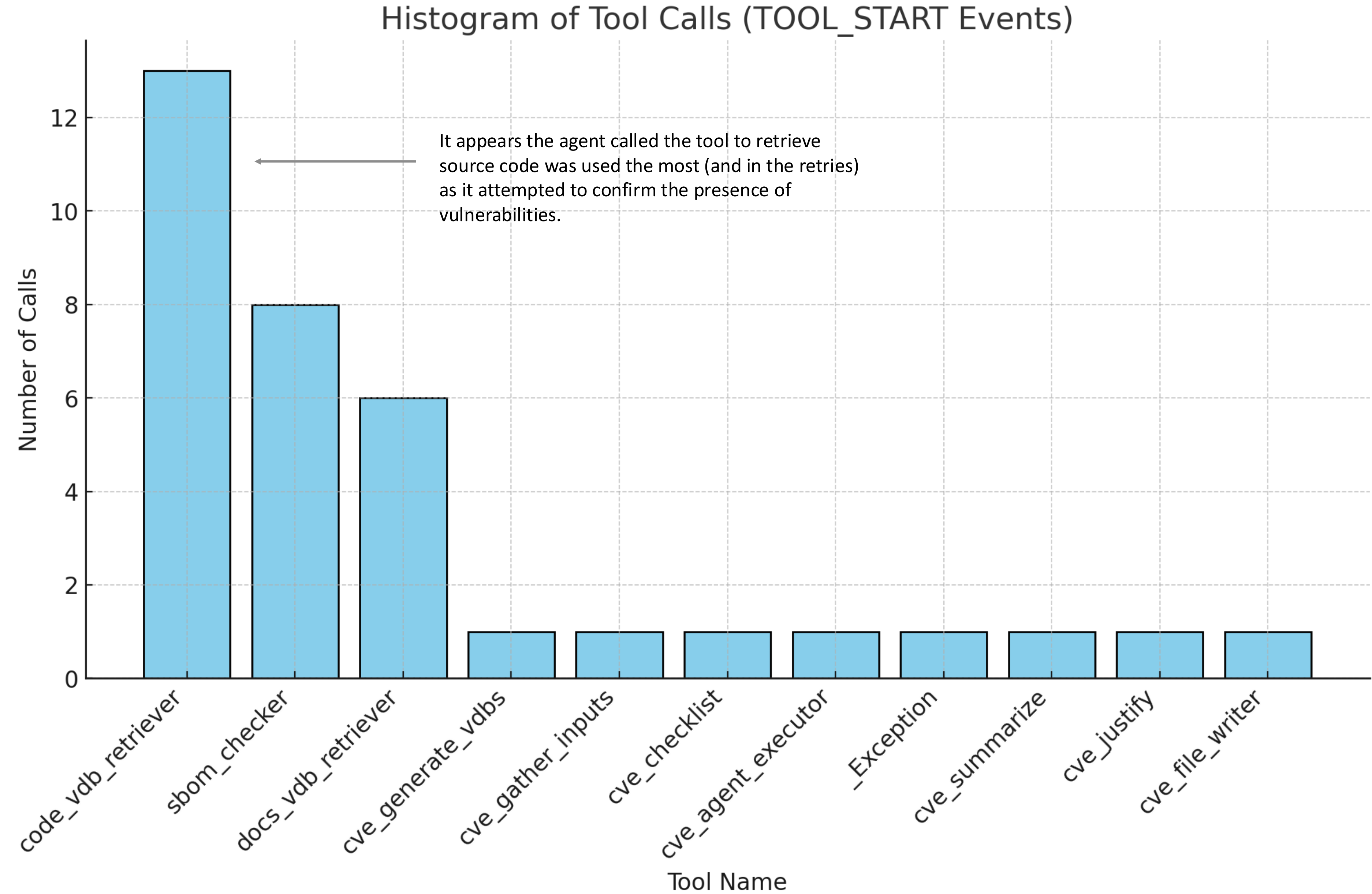
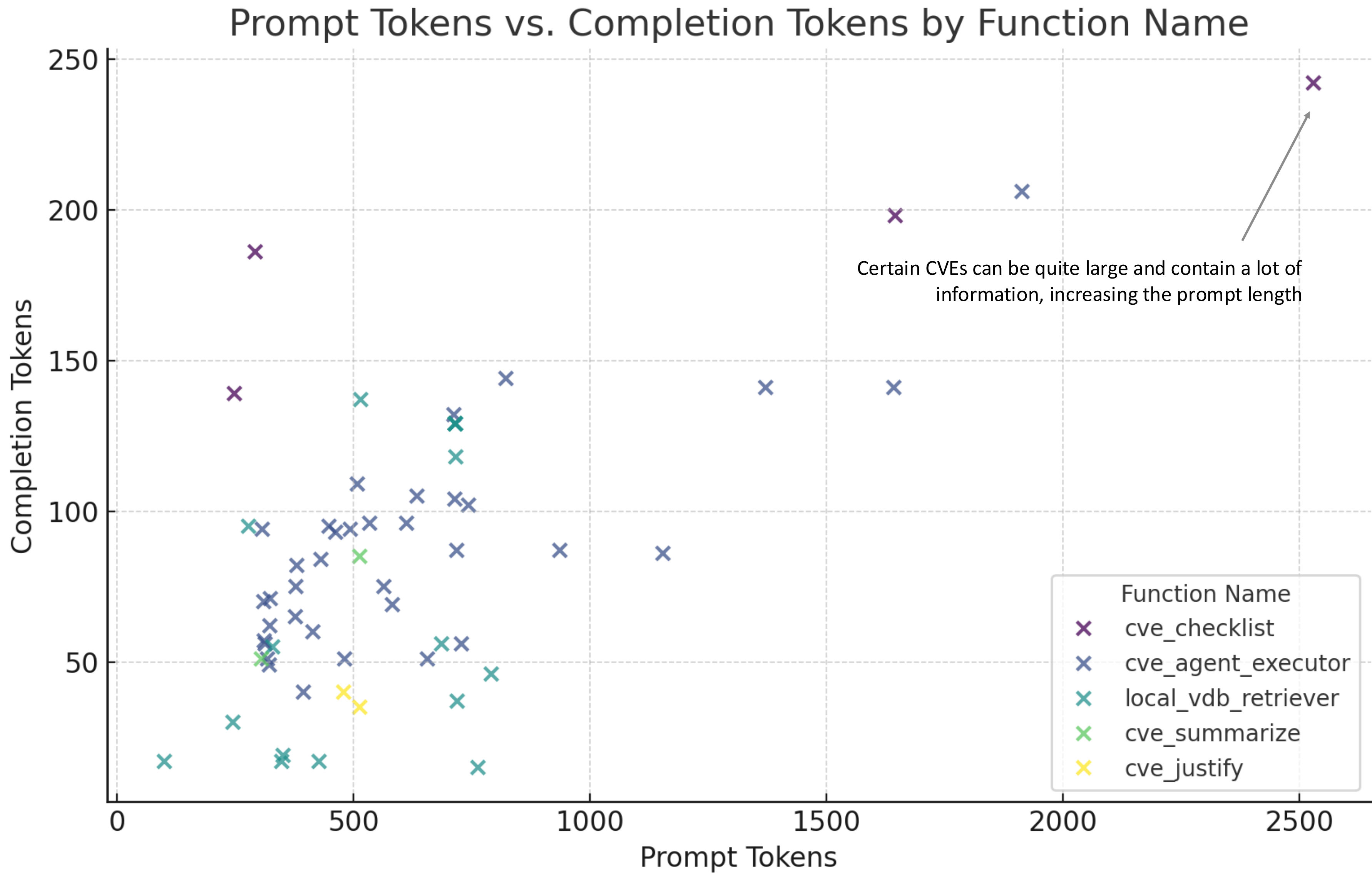
The AIQ Profiler also provides insight into LLM latency at various observed concurrencies, allowing us to optimize our LLM deployment strategy.



* All metrics are extracted from the AIQ profiling report, not pictured here for brevity.

Visualizing Other Behavior

The AIQ Profiler also collects fine-grained telemetry per function, tool, and LLM regardless of framework.



Visualized are just some examples of workflow behavior extracted from Profiler output.

AIQ Toolkit Component: Functions

Functions are generic coroutines wrapped in LLM frameworks at runtime by the WorkflowBuilder.
Everything is a Function!

```
class CurrentTimeToolConfig(FunctionBaseConfig, name="current_datetime"): 2 usages 1 Michael Demoret +1
    pass

@register_function(config_type=CurrentTimeToolConfig) 1 mpenn +1
async def current_datetime(config: CurrentTimeToolConfig, builder: Builder):

    import datetime

    async def _get_current_time(unused: str) -> str: 1 mpenn +1

        now = datetime.datetime.now() # Get current time
        now_human_readable = now.strftime("%Y-%m-%d %H:%M:%S")

        return f"The current time of day is {now_human_readable}" # Format time in H:MM AM/PM format

    yield FunctionInfo.from_fn(_get_current_time,
                               description="Returns the current date and time in human readable format.")
```

A typed Pydantic model used to provide a unique name to each function

Decorator to register this function with the toolkit by the unique name above + namespace

Each function is a coroutine with access to its configuration object and the WorkflowBuilder to construct any configured workflow/function/LLM/etc object

The function skill implementation

Yield a FunctionInfo object stored in the registration system and used by the WorkflowBuilder to construct framework wrapped coroutines at runtime

AIQ Toolkit workflow developers will use this interface to define new functions (tools) that can be accessed through the registration system by the application at runtime.

AIQ Toolkit Components: Workflows

Workflow instantiations serve as the execution manager for an AIQ Toolkit application

```
# Define the pipeline configuration
class AgentMemoryWorkflowConfig(FunctionBaseConfig, name="agent_memory"): 2 usages  ± mpenn +2
    verbose: bool = True

    tool_names: list[FunctionRef] = []
    llm: LLMRef
```

← A typed Pydantic model used to provide configuration settings and a unique name for each Function

```
@register_function(config_type=AgentMemoryWorkflowConfig, framework_wrappers=[LLMFrameworkEnum.LANGCHAIN]) ± dnandakur
async def agent_memory_workflow(config: AgentMemoryWorkflowConfig, builder: Builder):
    from langchain_core.messages import HumanMessage
    from langchain_core.messages import SystemMessage
    from langgraph.graph import START
    from langgraph.graph import MessagesState
    from langgraph.graph import StateGraph
    from langgraph.prebuilt import ToolNode
    from langgraph.prebuilt import tools_condition
```

← Decorator to register this Function by the unique name above

← Provides access to the Workflow configuration and WorkflowBuilder objects to provide access to all configured objects

```
# Use builder to generate framework specific tools and initialize the LLM
tools = builder.get_tools(tool_names=config.tool_names, wrapper_type=LLMFrameworkEnum.LANGCHAIN)
llm_n = await builder.get_llm(llm_name=config.llm, wrapper_type=LLMFrameworkEnum.LANGCHAIN)
```

← WorkflowBuilder constructs LLM framework wrapped tools by name(s)

← WorkflowBuilder constructs LLM frameworks wrapped clients by name(s)

```
# This function will be called with the input message
async def _response_fn(input_message: str) -> str: ± dnandakumar-nv +1
```

← Define the main coroutine that processes incoming messages.

```
    try:
        input_message = "[user_id: john_doe] " + input_message
        logger.debug("Starting agent execution")

        output = await agent_executor.ainvoke(input={"messages": [HumanMessage(content=input_message)]})

        return output["messages"][-1].content
    finally:
        logger.debug("Finished agent execution")
```

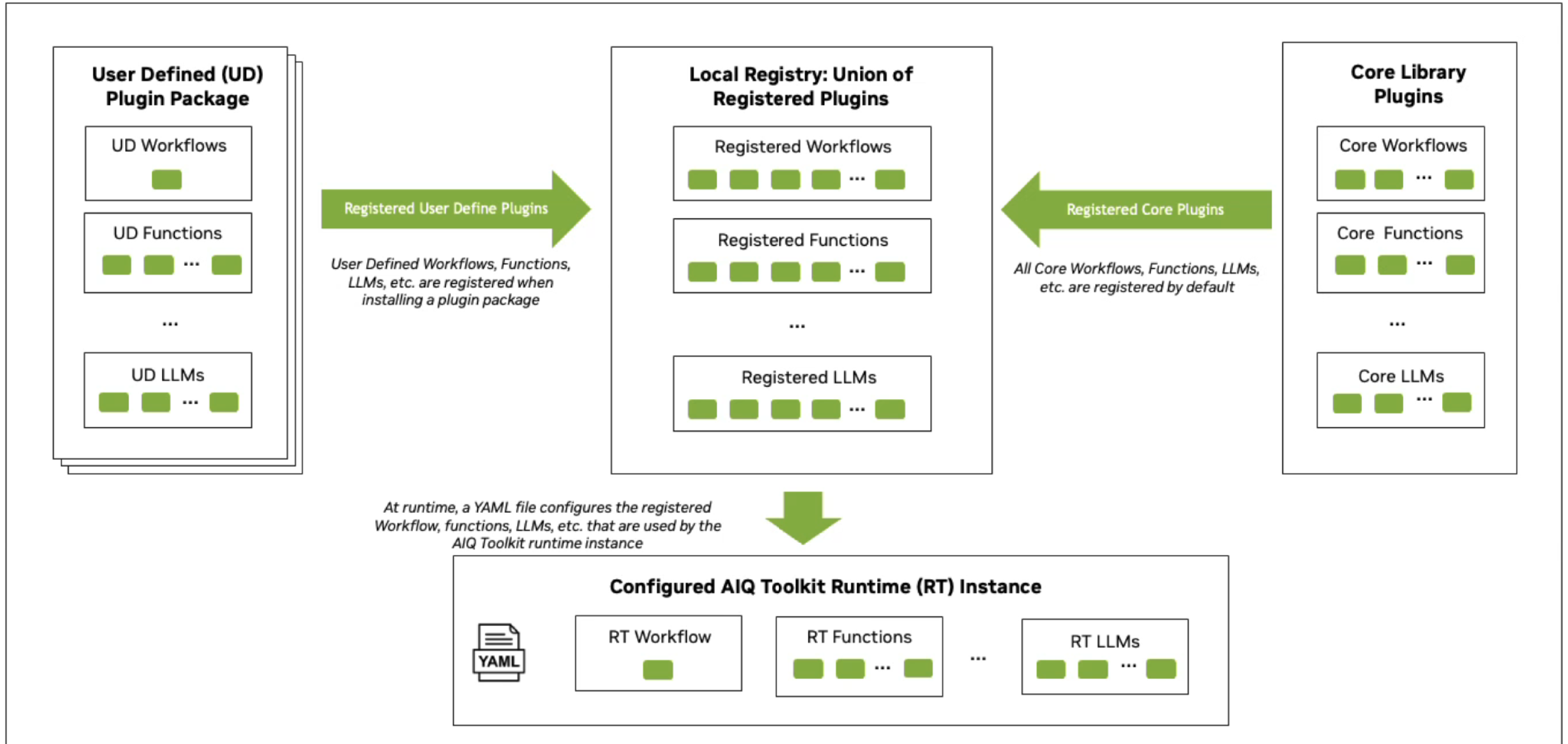
```
try:
    yield FunctionInfo.create(single_fn=_response_fn)
except GeneratorExit:
    logger.exception(msg="Exited early!", exc_info=True)
finally:
    logger.debug("Cleaning up")
```

← yield main coroutine that will be used to process incoming messages and perform any necessary clean up upon exit

Notice workflows are also registered Functions!

The AIQ Toolkit Registration System

The registration systems allows developers build reusable components of an agentic system



Unified Agentic AI: Collaboration, Communication, and Tool Access

NVIDIA AIQ Toolkit, Anthropic MCP, and Google A2A

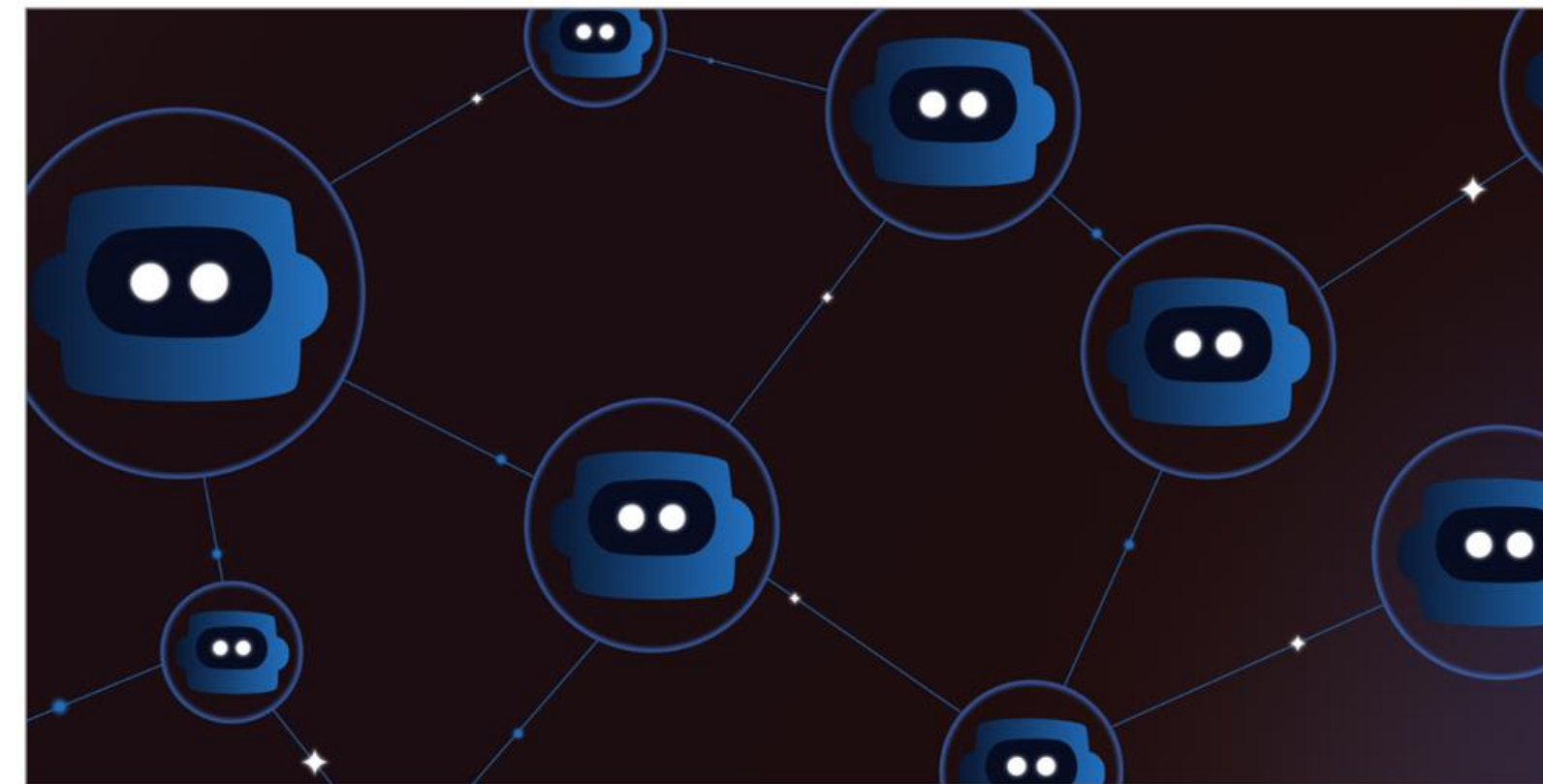
NVIDIA AIQ Toolkit



Optimization, profiling, and framework agnostic connections

- Comprehensive toolkit for building, profiling, and optimizing teams of AI agents across any framework
- Provides universal descriptors for agents and tools, enabling cross-framework compatibility
- Focuses on performance optimization, evaluation, and observability of entire agentic systems

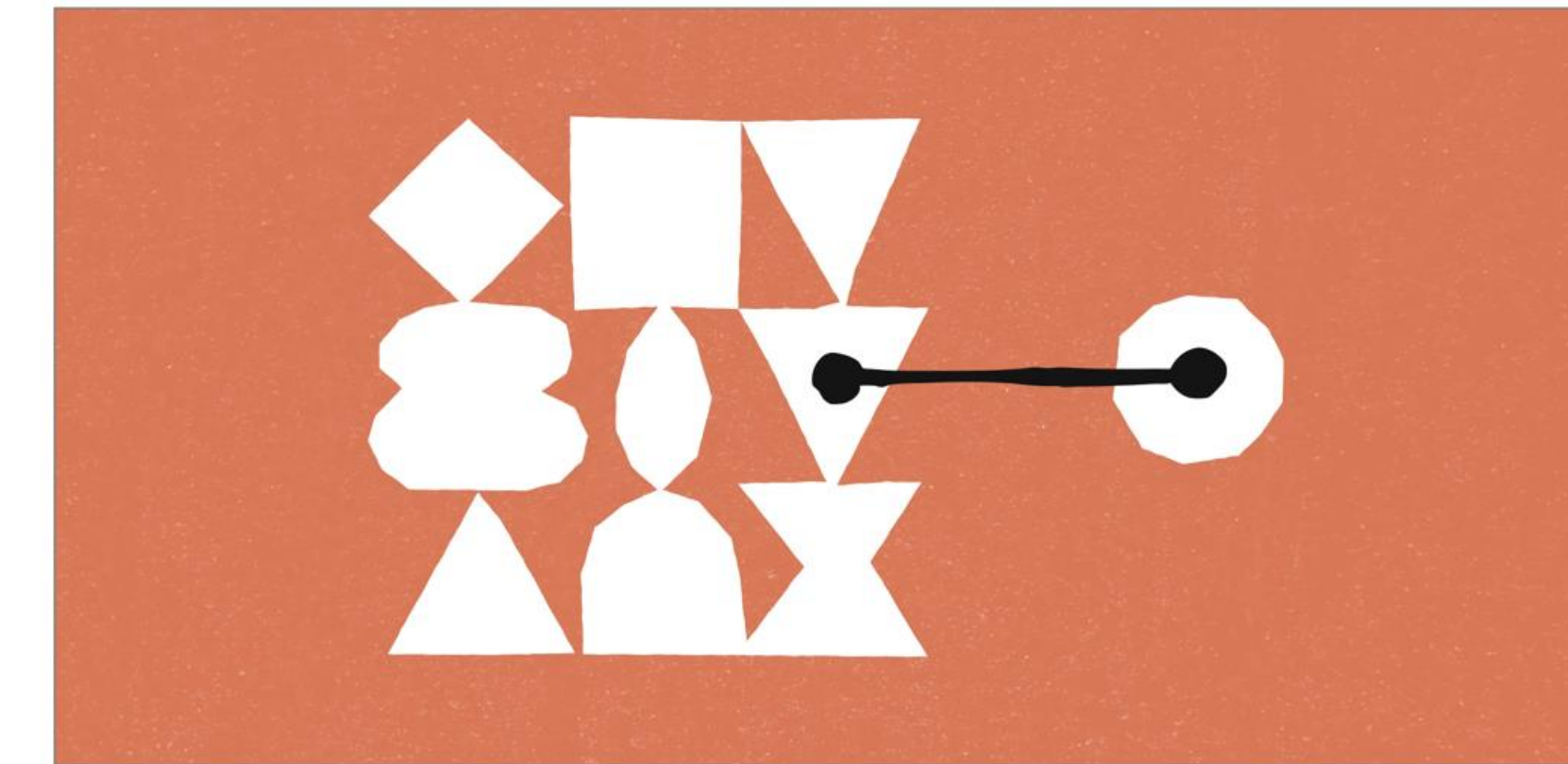
MCP



Standardized agent-to-tool integration

- Focuses on vertical integration—connecting individual AI models with external tools and data
- Optimized for direct request-response flows between a model and its tools
- Designed for controlled tool access rather than peer-to-peer agent collaboration

A2A



Standardized agent-to-agent communication

- Focuses on horizontal integration—enabling communication between different AI agents
- Creates standardized protocols for agents to discover each other and collaborate across platforms
- Designed primarily for agent-to-agent communication rather than tool usage or optimization

NVIDIA Blueprints for Building AI Agents

An Easy Starting Point for Building Fast, Smart, Enterprise-Grade AI Agents

NVIDIA AI Blueprints

PDF to Podcast

AI Assistant for Customer Service

Vulnerability Analysis for Container Security

Generative Virtual Screening for Drug Discovery

Video Search and Summarization

Specialized AI Agents



Research Assistant Agent



Customer Service Agent



Software Security Agent



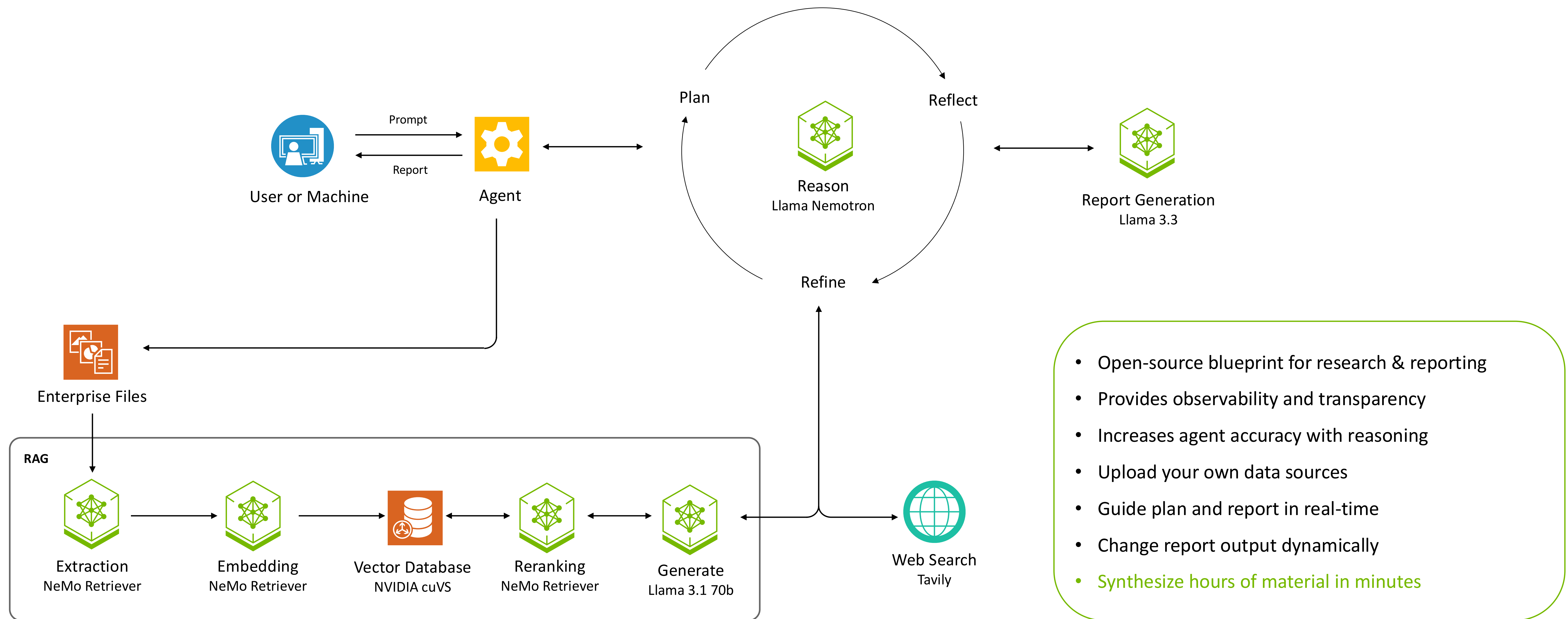
Virtual Lab Agent



Video Analytics Agent

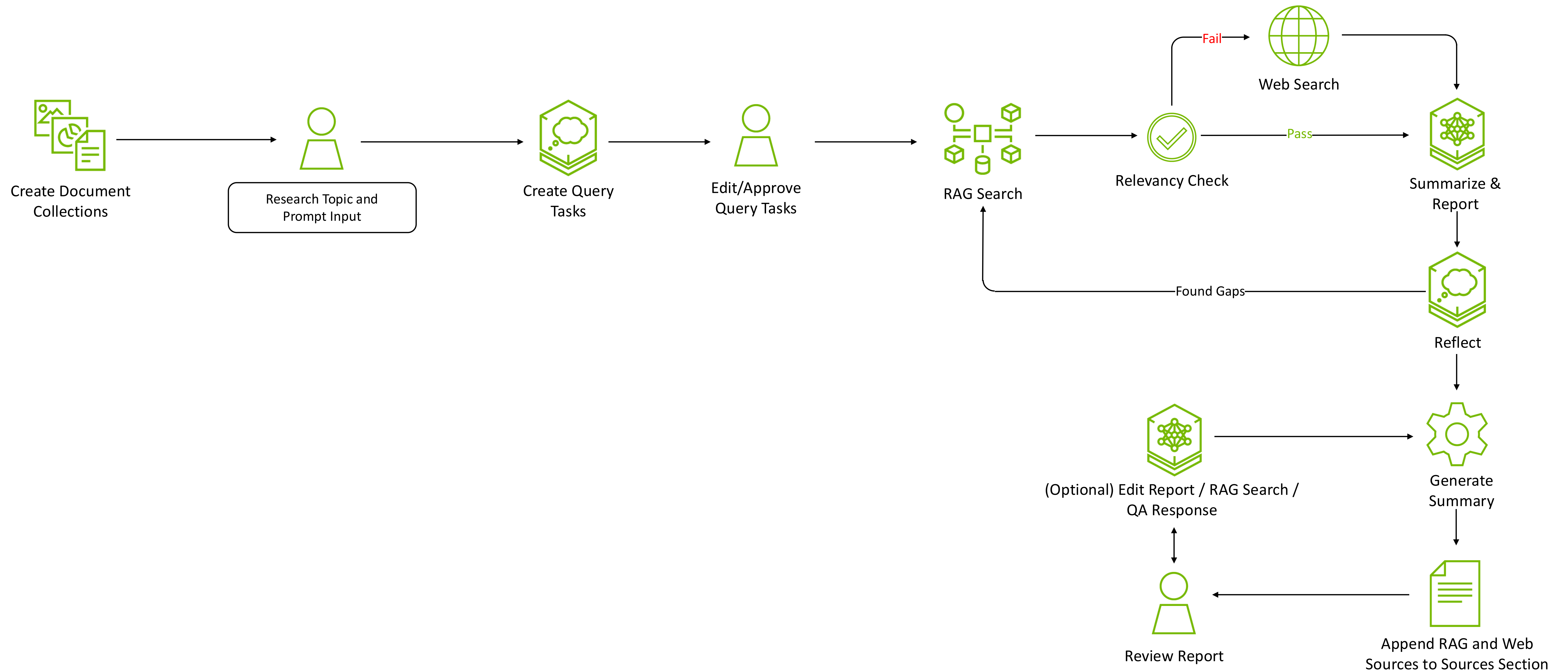
AIQ (Pronounced 'IQ') NVIDIA Blueprint

Use Reasoning to Connect AI Agents, Data, and Tools



AI-Q Research Assistant Blueprint Process Flow

Deep Research with On-Premise Data: Chat with Your Enterprise Data

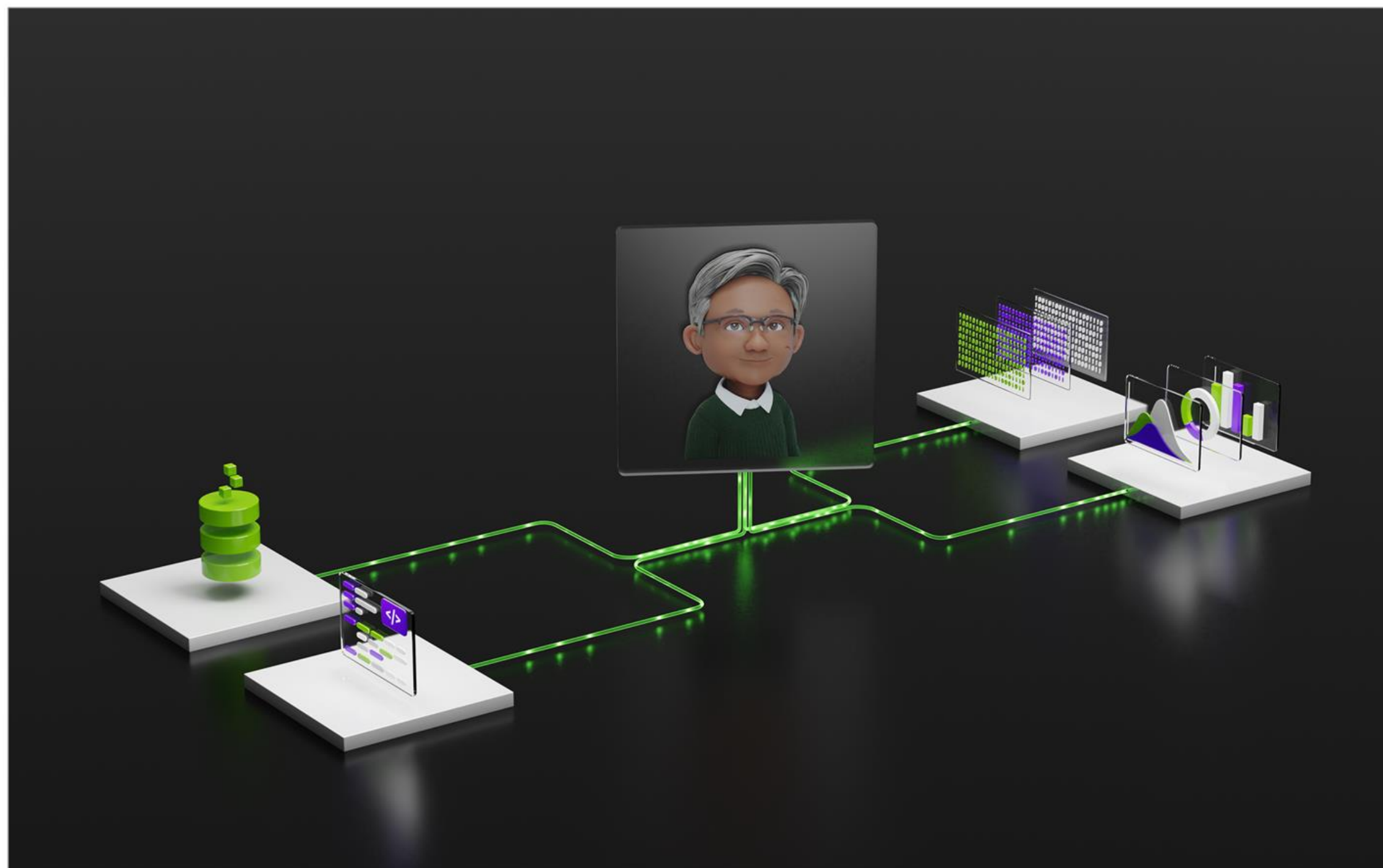


Building Blocks of the AI-Q NVIDIA Blueprint

Use Reasoning to Connect AI Agents, Data, and Tools

Llama Nemotron

Equips AI Agents with Reasoning Skills



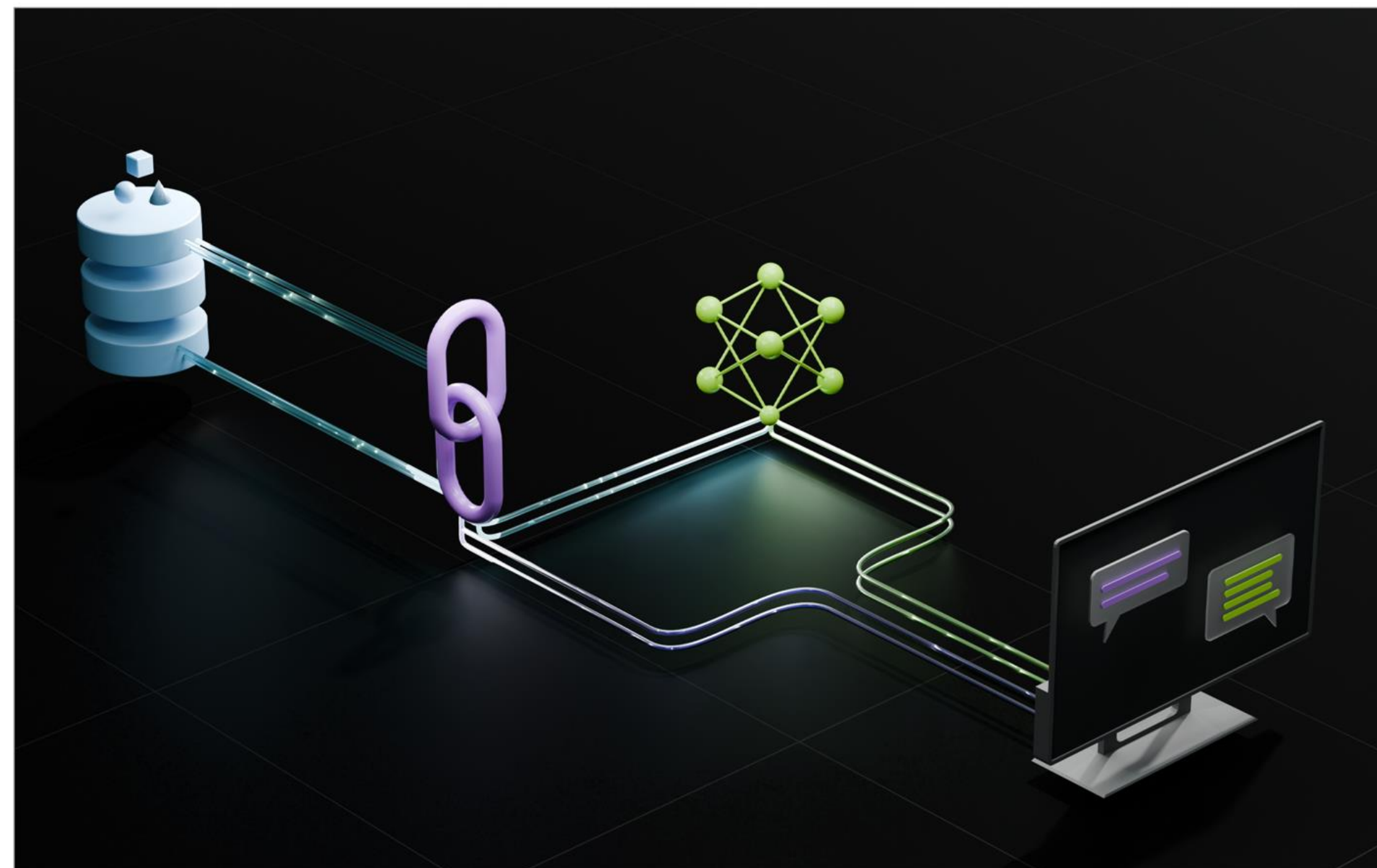
High accuracy and throughput
on a single GPU

#1

Open Reasoning Models

NeMo Retriever

Connects AI Agents to Data



Fast, accurate, multimodal
data processing at scale

15X Faster

Data Extraction

Agent Intelligence (AIQ) Toolkit

Monitor, Evaluate, and Optimize Agentic Systems



Find hidden bottlenecks and
costs to scale agentic systems

50% Greater

Retrieval Accuracy

NVIDIA Technologies for Building Agentic AI Applications

Included with NVIDIA AI Enterprise

Learning & Adaptation

Continuous feedback and improvements



NVIDIA NeMo

Safety, Security & Privacy

Preventing unwanted behaviors and protecting proprietary data



NVIDIA NeMo Guardrails

Reliability of Responses

Best accuracy and throughput at scale



NVIDIA NeMo Retriever

User-AI Interaction

Natural, effective communications and understanding



NVIDIA Digital Humans

Llama Nemotron Reasoning Model Family

Leading Open Reasoning NIM Microservices for Agentic AI

Nano



Super



Ultra



NVIDIA NeMo
Framework

Post-Training

Open NVIDIA
Datasets

Model-Building
Recipe

60B Tokens, 360K H100 Hours, 45K Annotation Hours



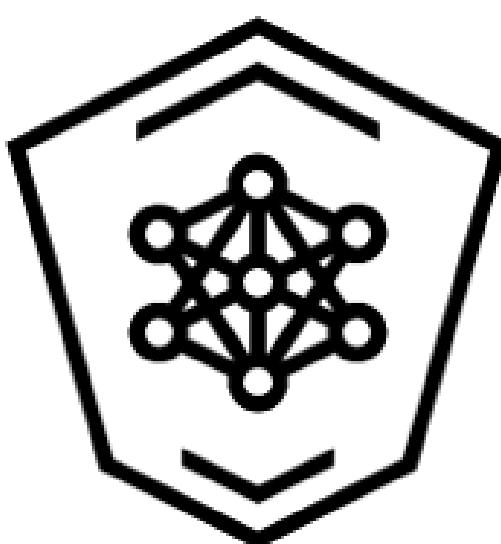
Leading Accuracy



Highest Efficiency



Reasoning ON/OFF



Enterprise Ready

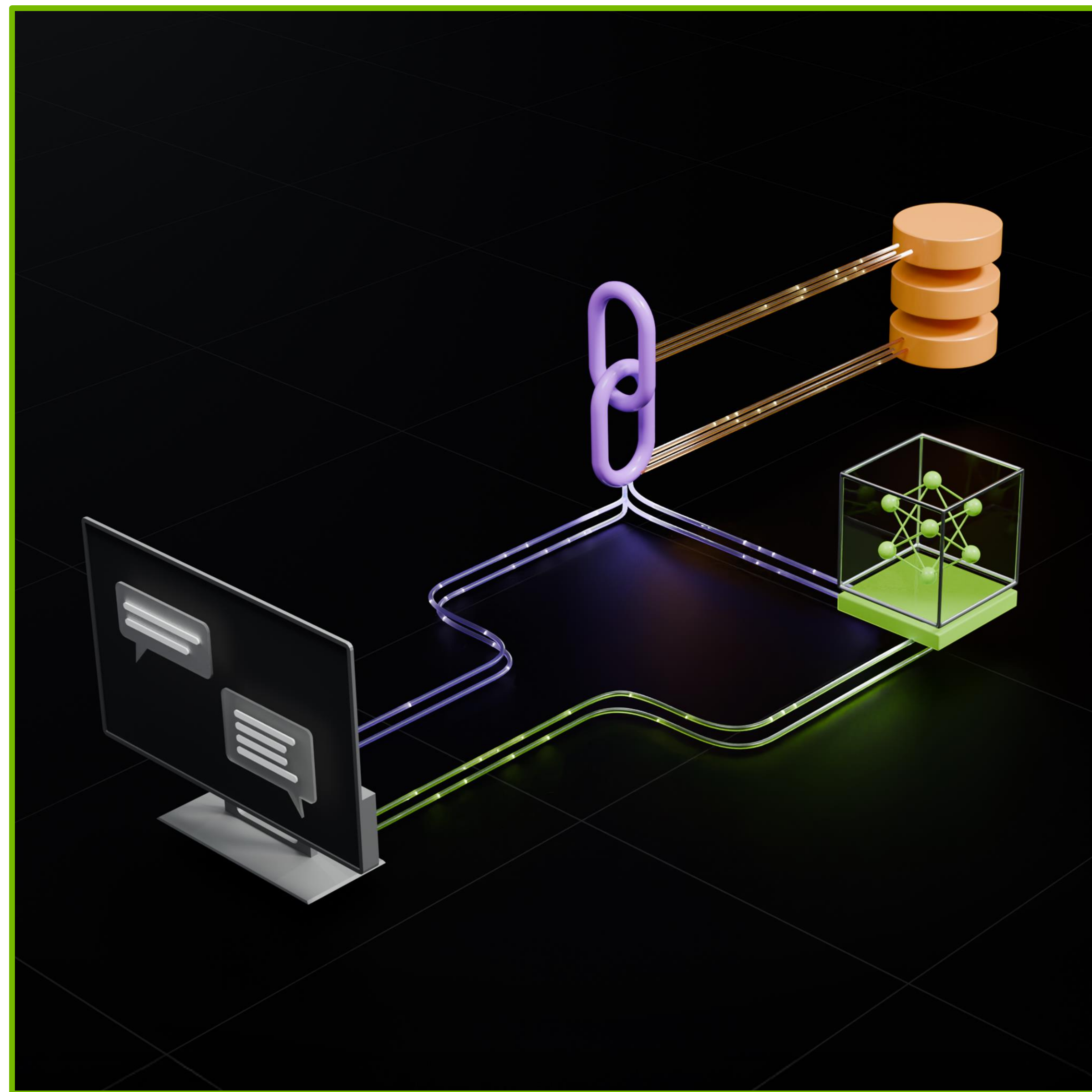


Open

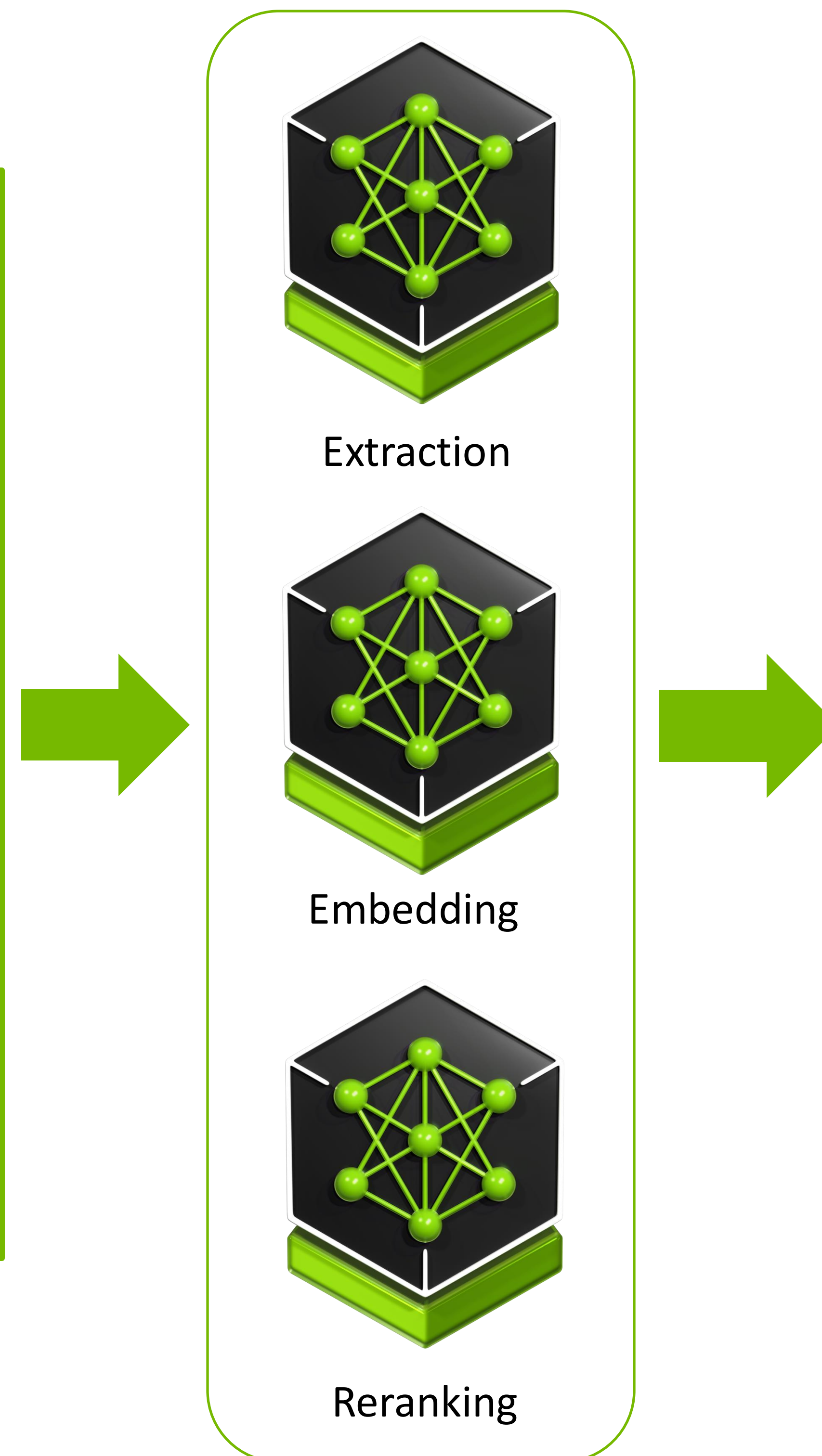
NVIDIA NeMo Retriever: Accurate, Real Time Insights from Enterprise Data

Now Supports Even More Data Types – Including Text, Tables, Charts, and Infographics from Millions of PDFs

NVIDIA AI Blueprint for RAG

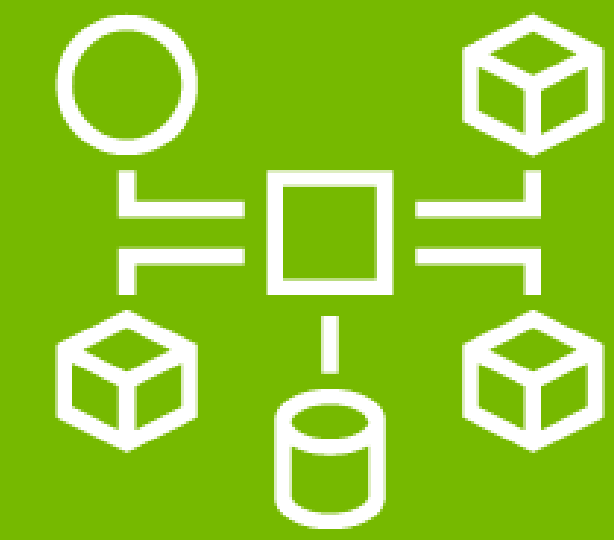


NeMo Retriever



50%

fewer incorrect
answers



15x

improved multimodal
extraction throughput



35x

improved data
storage efficiency



20%

reduced
RAG costs

Developed With AI



AI Research Assistant
Agent



Virtual Lab AI
Agent



Software Security AI
Agent



Video Analytics AI
Agent



AI Sales Development
Representative Agent



Customer Service AI
Agent



Financial Analyst AI
Agent



Employee Support AI
Agent



Factory Operations AI
Agent

[NVIDIA AI Building Blocks for Agentic AI](#)

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- **Building LLM Applications With Prompt Engineering**
- **Deploying a Model for Inference at Production Scale**
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- **Synthetic Data Generation for Training Computer Vision Models**
- **Techniques for Improving the Effectiveness of RAG Systems**

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Software

100s of APIs, models, SDKs, microservices, and early access to NVIDIA tech

Training

Hands-on self-paced courses, instructor-led workshops, and certifications

GPU Sandbox

Approval basis, multi-GPU and multi-node

Learning

Tutorials, self-paced courses, blogs, documentation, code samples

Community

Dedicated developer forums, meetups, hackathons

Ecosystem

GTC, NVIDIA Partner Network

Organizations

Startups

Cloud credits, engineering resources, technology discounts, exposure to VCs

Venture Capital

Deal flow and portfolio support for Venture Capital firms

Higher Education

Teaching kits, training, curriculum co-development, grants

ISVs and SIs

Engineering guidance, discounts, marketing opportunities

Research

Grant programs, collaboration opportunities

Enterprises

Tailored developer training, skills certification, technical support



Thank You !

GTC 2025- Watch
on Demand



What's Next in AI
Starts Here

March 17-21, 2025

